

# The problem: exams that cannot be trusted

Saakshi is a computer exam system where every answer is signed, every failure is survived and every decision can be checked.

## What really happened in India

**22.79 lakh**

NEET-UG 2026 candidates,  
exam cancelled

**45 hrs**

a guess paper circulated  
before the exam

**~120 of  
410**

guess paper questions  
matched the real paper

**21 June**

re-exam, 46 days after  
cancellation

**3 May**

Exam held

**12 May**

Cancelled

**21 June**

Re-exam, no fee

**25 July**

Education Minister resigns

**From 2027**

NEET moves to CBT

### NEET-UG 2026: a leak at the source

Insiders at a printing press scanned papers before dispatch. The leak spread to more than 10 states. The CBI made arrests. The leak came from the people who handle the paper, not from the exam hall.

### NEET-UG 2024: unfair fixes

1,563 candidates got grace marks for lost time. The marks were scrapped and a re-test was offered. Only 813 took it. The court noted wrong paper sets and weak CCTV.

### SSC 2025: the insider at the centre

Exams were cancelled after crashes and biometric failures. At Dhanbad the server manager was arrested. Answers were auto-selected while candidates only moved the mouse.

### JEE Main 2025: no proof for the candidate

A pop-up showed 46 questions attempted. The response sheet showed 29. Candidates can challenge the answer key, but not their recorded responses.

**Why now:** about 22 lakh candidates will sit NEET on computers from 2027. The system they use must be ready before that.

Sources: Wikipedia "2026 NEET controversy"; Outlook India (NEET-UG 2026 cancelled, re-exam 21 June; Education Minister resigns); Gulf News and SCC Online (guess paper, 45 hours); MedicalDialogues (NEET CBT from 2027); SCC Online 2 Aug 2024 (NEET-UG 2024); Careers360 (SSC Phase 13, SSC CGL Dhanbad, JEE Main 2025); Bihar CHO exam report, All India Radio news. Full list in docs/research.md.

# The problem, broken into six failure types

Each failure type needs a different answer. One fix does not cover all of them.



### 1. Paper leak at the source

**Seen in:** NEET-UG 2026

Someone who sets or prints the paper shares it.

Saakshi: papers stay locked until two of three key holders release them. It cannot stop a setter who knows the questions.



### 2. Insider changes answers

**Seen in:** SSC Dhanbad, Bihar CHO (37 arrests)

A centre server or its operator edits or injects responses.

Saakshi: each answer is signed on the seat and chained. Any later edit is found and the true answer is recovered.



### 3. Outage and crashes

**Seen in:** CUET-UG 2026 (3,765 candidates re-tested), SSC 2025

Software or links fail during the exam.

Saakshi: the seat keeps answers on its own disk. Killed relays and cells were rebuilt with 0 answers lost.



### 4. Remote help and cheating tools

**Seen in:** SSC 2025 remote access, Bihar CHO proxy server

Remote-view software or hidden tools run on the exam PC.

Saakshi: a check before the exam blocks known remote tools by name. Flags go to a human, never an automatic penalty.



### 5. No proof for the candidate

**Seen in:** JEE Main 2025 (46 vs 29)

The candidate cannot check what was recorded.

Saakshi: each candidate gets a receipt slip. The public verify page compares it with the record.



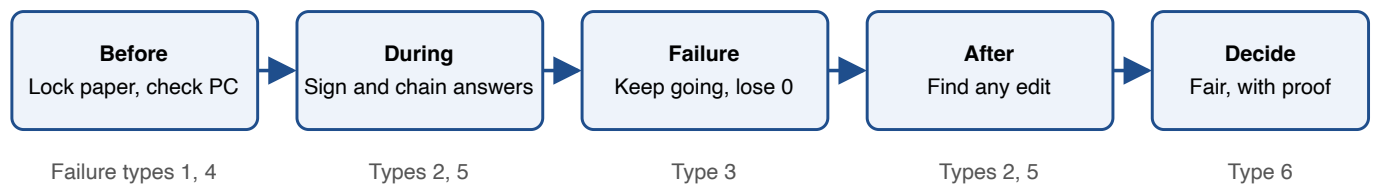
### 6. Unfair handling after trouble

**Seen in:** NEET-UG 2024 (grace marks, 813 of 1,563 re-tested)

Who gets compensated, re-tested or re-conducted is argued in court.

Saakshi: a rule-based decision per candidate and per centre, shown with evidence. Tested on synthetic data only.

## How Saakshi answers them, in one picture



Incidents: Careers360 (SSC, JEE Main 2025); Business Standard (CUET-UG 2026 re-test); All India Radio (Bihar CHO); SCC Online (NEET-UG 2024). Saakshi results: do cs/evidence/ (chaos drill 20 of 20 passed, 16,000 answers sent and 16,000 stored; tamper lab found 1,000 of 1,000 byte flips).

## Does the solution fit the problem? The 12 focus areas

**7 of 12**

built and tested

**5 of 12**

partly built, rest named below

**0 of 12**

not started

| #  | Focus area                         | What Saakshi does                                                                                                                       | Fit |
|----|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-----|
| 1  | Real-time monitoring               | Live seat grid and control-room tiles per centre. A silent seat shows after 30 seconds.                                                 | ~   |
| 2  | Early failure prediction           | Risk score per centre: allot, add an observer, or do not allot, with the top 3 reasons. Synthetic drill data only.                      | ~   |
| 3  | Incident detection and escalation  | Incidents ranked by severity, with an escalation ladder, a regulator report draft and invigilator reports sorted by type.               | ✓   |
| 4  | Backup and disaster recovery       | Relays store and forward. A wiped cell is rebuilt from relays. 3 cells run together. Lost answers: 0.                                   | ~   |
| 5  | Tamper-evident storage             | Signed, chained answers. A per-shift register that an independent witness checks. Keys split 2 of 3.                                    | ✓   |
| 6  | Suspicious patterns                | Pattern signals on synthetic data: 53 flags, no honest candidate flagged. A human always reviews.                                       | ~   |
| 7  | Reconciliation and validation      | Missing answers are requested again. Duplicates are ignored. Each centre-shift report ties registered, submitted and receipts together. | ✓   |
| 8  | Candidate communication            | Tick states, offline message and receipt slip in English and Hindi. Approved notices in English, Hindi and Tamil.                       | ✓   |
| 9  | Re-schedule or re-conduct decision | Per-candidate compensate or re-test, plus centre-level re-conduct. Checked on NEET-UG 2024 and CUET 2026 cases.                         | ✓   |
| 10 | Fairness when disrupted            | Rules that treat candidates alike. Time-gap records and caps on compensation are still to build.                                        | ~   |
| 11 | Audit trail and evidence reports   | A signed evidence pack anyone can verify. The verify page shows the exact changed answer.                                               | ✓   |
| 12 | AI analytics for systemic risk     | Risk scores, incident sorting and pattern signals on the cells' own export.                                                             | ✓   |

✓ built and tested ~ partly built

### What the solution does well

It removes the need to trust any one insider. It keeps the exam running through failures. It gives the candidate proof.

### Honest limits

A seat signature shows a record was not changed after the device made it. It does not show the candidate chose it. It cannot stop a paper setter from leaking questions. Analytics use synthetic data. Speed figures come from one machine. Tested on macOS and Windows.

Repository: <https://github.com/anirudh12032008/saakshi>